STATIC TIMING ANALYSIS

SKY130 HD · 16 PVT corners · post-route path-list review

Purpose

Execution record: OpenROAD static timing analysis was run for all 16 PVT corners on the routed design. Inputs included 6_1_fill.odb, 6_1_fill.sdc and 6_final.spef. Per-corner logs were generated in reports/sta/, with setup- and hold-timing report sections. This document summarizes those reported results.

16	10	6	-32.48 ns
Corners reported	Setup status MET	Setup violations	Worst reported WNS*

Key findings

• 10 of 16 corners are marked MET and 6 are marked VIOLATED in the supplied summary.
• The lowest reported setup slack is -32.48 ns at ss_n40C_1v28 (-40 °C, 1.28 V).
• The strongest reported setup slack is 7.95 ns at ff_100C_1v95.
• All reported minimum hold slacks are positive; the listed violations are setup-timing violations, not reported hold violations.

Metric interpretation

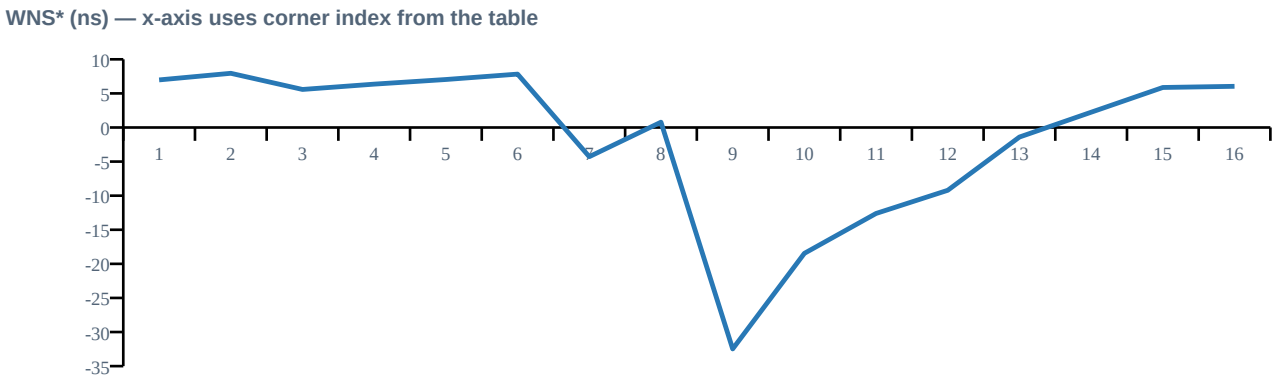
WNS* is the minimum setup slack among the setup paths printed in each supplied log. TNS* is the sum of negative setup slacks among those printed paths only. These are path-list-derived values, not guaranteed design-wide WNS/TNS. A positive hold value is the minimum hold slack reported in the supplied summary.

All-corner results

Table 1. Reported metrics across the 16 PVT corners.

Corner	Proc.	Temp °C	V	WNS* ns	TNS* ns	Hold min ns	Neg. setup paths	Status
ff_100C_1v65	FF	100	1.65	6.98	0.00	0.25	0 / 100	MET
ff_100C_1v95	FF	100	1.95	7.95	0.00	0.20	0 / 100	MET
ff_n40C_1v56	FF	-40	1.56	5.58	0.00	0.27	0 / 100	MET
ff_n40C_1v65	FF	-40	1.65	6.36	0.00	0.25	0 / 100	MET
ff_n40C_1v76	FF	-40	1.76	7.04	0.00	0.22	0 / 100	MET
ff_n40C_1v95	FF	-40	1.95	7.83	0.00	0.19	0 / 100	MET
ss_100C_1v40	SS	100	1.40	-4.28	-319.73	0.86	100 / 100	VIOLATED
ss_100C_1v60	SS	100	1.60	0.78	0.00	0.61	0 / 100	MET
ss_n40C_1v28	SS	-40	1.28	-32.48	-2840.22	1.39	100 / 100	VIOLATED
ss_n40C_1v35	SS	-40	1.35	-18.46	-1585.59	1.08	100 / 100	VIOLATED
ss_n40C_1v40	SS	-40	1.40	-12.61	-1054.61	0.92	100 / 100	VIOLATED
ss_n40C_1v44	SS	-40	1.44	-9.21	-744.48	0.82	100 / 100	VIOLATED
ss_n40C_1v60	SS	-40	1.60	-1.40	-46.33	0.57	63 / 100	VIOLATED
ss_n40C_1v76	SS	-40	1.76	2.24	0.00	0.44	0 / 100	MET
tt_025C_1v80	TT	25	1.80	5.87	0.00	0.30	0 / 100	MET
tt_100C_1v80	TT	100	1.80	6.04	0.00	0.31	0 / 100	MET

Graph 1. Reported setup WNS* across corners



Corner index follows the table order. Negative WNS* indicates that at least one setup path in the reported set misses timing.

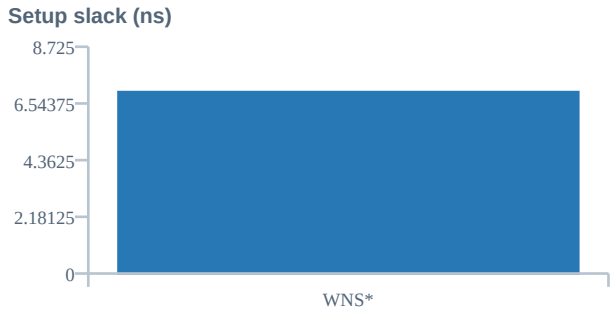
ff_100C_1v65

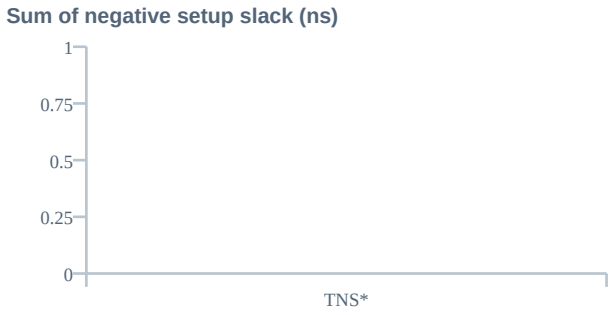
Process: FF | Temperature: 100 °C | Voltage: 1.65 V

MET

Metric	Reported value	Interpretation
WNS* / minimum reported setup slack	6.9800 ns	Non-negative for the reported setup path set.
TNS* / sum of negative reported setup slacks	0.0000 ns	Zero means no negative setup slack was summed in the reported set.
Minimum reported hold slack	0.2500 ns	Positive in the supplied summary; no negative hold paths are reported.
Negative setup paths in reported set	0 of 100	Count copied from the supplied summary.
Negative hold paths in reported set	0 of 100	Count copied from the supplied summary.

Corner metrics





Assessment

The supplied summary marks setup timing MET for the reported path set and shows positive minimum hold slack. This is not a substitute for confirming complete design-wide STA summaries.

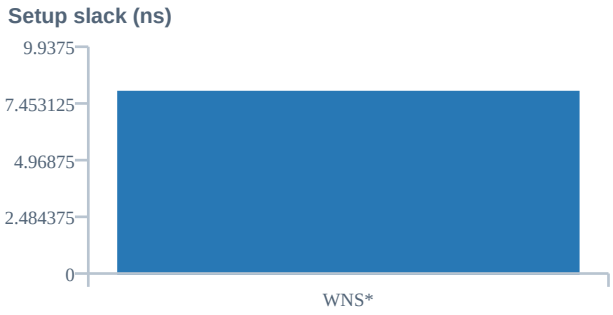
ff_100C_1v95

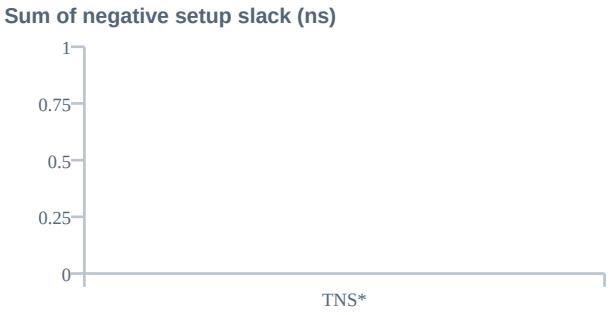
Process: FF | Temperature: 100 °C | Voltage: 1.95 V

MET

Metric	Reported value	Interpretation
WNS* / minimum reported setup slack	7.9500 ns	Non-negative for the reported setup path set.
TNS* / sum of negative reported setup slacks	0.0000 ns	Zero means no negative setup slack was summed in the reported set.
Minimum reported hold slack	0.2000 ns	Positive in the supplied summary; no negative hold paths are reported.
Negative setup paths in reported set	0 of 100	Count copied from the supplied summary.
Negative hold paths in reported set	0 of 100	Count copied from the supplied summary.

Corner metrics





Assessment

The supplied summary marks setup timing MET for the reported path set and shows positive minimum hold slack. This is not a substitute for confirming complete design-wide STA summaries.

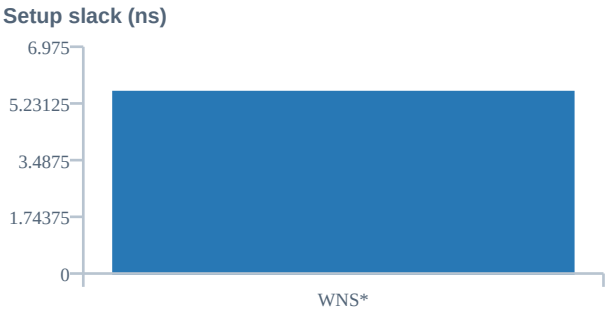
ff_n40C_1v56

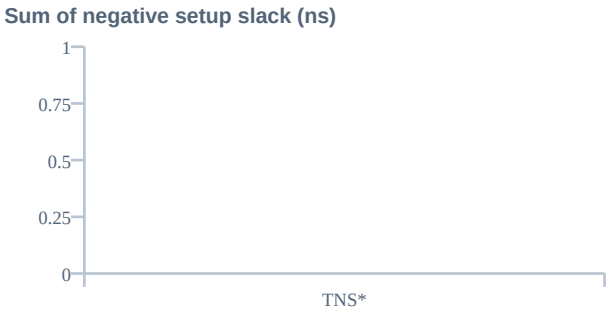
Process: FF | Temperature: -40 °C | Voltage: 1.56 V

MET

Metric	Reported value	Interpretation
WNS* / minimum reported setup slack	5.5800 ns	Non-negative for the reported setup path set.
TNS* / sum of negative reported setup slacks	0.0000 ns	Zero means no negative setup slack was summed in the reported set.
Minimum reported hold slack	0.2700 ns	Positive in the supplied summary; no negative hold paths are reported.
Negative setup paths in reported set	0 of 100	Count copied from the supplied summary.
Negative hold paths in reported set	0 of 100	Count copied from the supplied summary.

Corner metrics





Assessment

The supplied summary marks setup timing MET for the reported path set and shows positive minimum hold slack. This is not a substitute for confirming complete design-wide STA summaries.

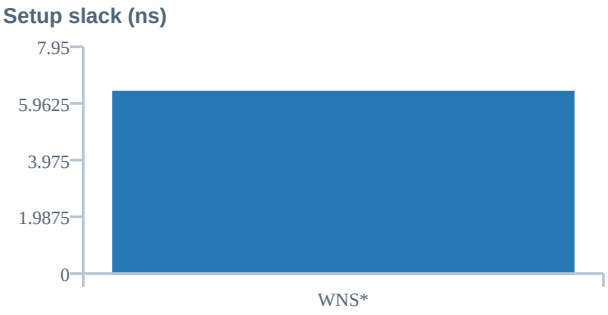
ff_n40C_1v65

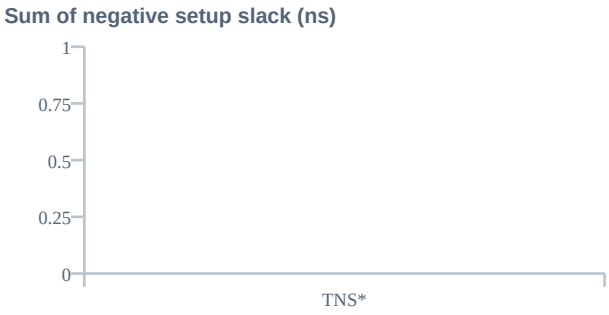
Process: FF | Temperature: -40 °C | Voltage: 1.65 V

MET

Metric	Reported value	Interpretation
WNS* / minimum reported setup slack	6.3600 ns	Non-negative for the reported setup path set.
TNS* / sum of negative reported setup slacks	0.0000 ns	Zero means no negative setup slack was summed in the reported set.
Minimum reported hold slack	0.2500 ns	Positive in the supplied summary; no negative hold paths are reported.
Negative setup paths in reported set	0 of 100	Count copied from the supplied summary.
Negative hold paths in reported set	0 of 100	Count copied from the supplied summary.

Corner metrics





Assessment

The supplied summary marks setup timing MET for the reported path set and shows positive minimum hold slack. This is not a substitute for confirming complete design-wide STA summaries.

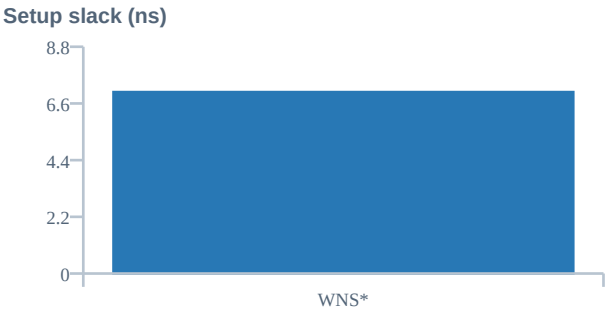
ff_n40C_1v76

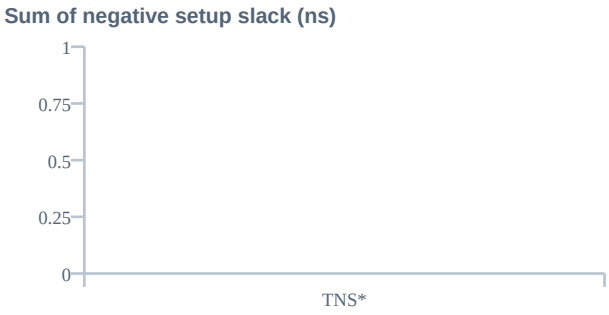
Process: FF | Temperature: -40 °C | Voltage: 1.76 V

MET

Metric	Reported value	Interpretation
WNS* / minimum reported setup slack	7.0400 ns	Non-negative for the reported setup path set.
TNS* / sum of negative reported setup slacks	0.0000 ns	Zero means no negative setup slack was summed in the reported set.
Minimum reported hold slack	0.2200 ns	Positive in the supplied summary; no negative hold paths are reported.
Negative setup paths in reported set	0 of 100	Count copied from the supplied summary.
Negative hold paths in reported set	0 of 100	Count copied from the supplied summary.

Corner metrics





Assessment

The supplied summary marks setup timing MET for the reported path set and shows positive minimum hold slack. This is not a substitute for confirming complete design-wide STA summaries.

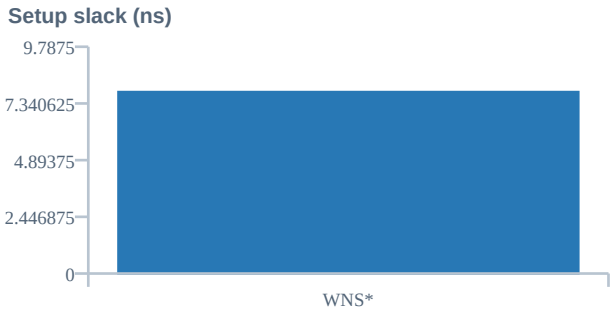
ff_n40C_1v95

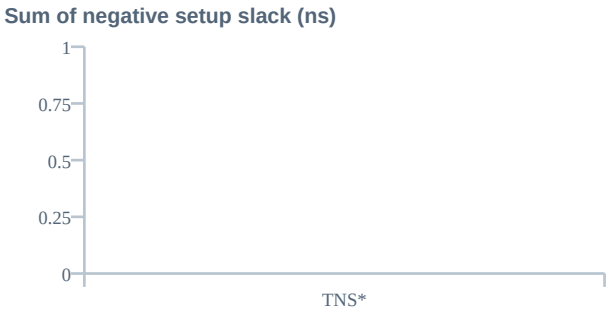
Process: FF | Temperature: -40 °C | Voltage: 1.95 V

MET

Metric	Reported value	Interpretation
WNS* / minimum reported setup slack	7.8300 ns	Non-negative for the reported setup path set.
TNS* / sum of negative reported setup slacks	0.0000 ns	Zero means no negative setup slack was summed in the reported set.
Minimum reported hold slack	0.1900 ns	Positive in the supplied summary; no negative hold paths are reported.
Negative setup paths in reported set	0 of 100	Count copied from the supplied summary.
Negative hold paths in reported set	0 of 100	Count copied from the supplied summary.

Corner metrics





Assessment

The supplied summary marks setup timing MET for the reported path set and shows positive minimum hold slack. This is not a substitute for confirming complete design-wide STA summaries.

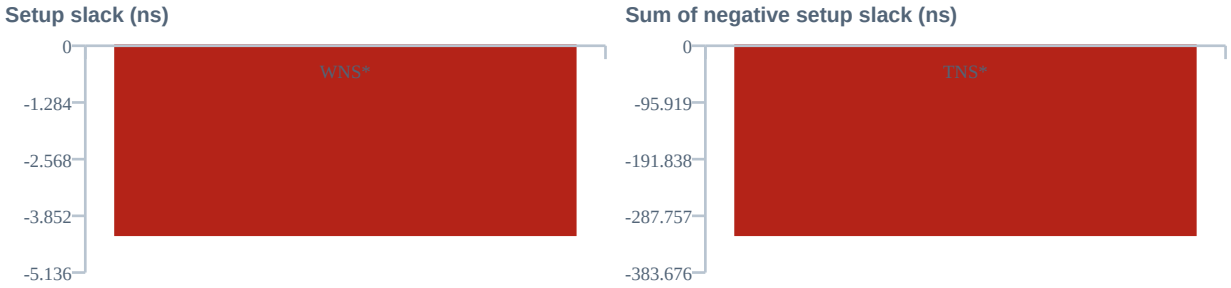
ss_100C_1v40

Process: SS | Temperature: 100 °C | Voltage: 1.40 V

VIOLATED

Metric	Reported value	Interpretation
WNS* / minimum reported setup slack	-4.2800 ns	Negative means a setup violation exists in the reported path set.
TNS* / sum of negative reported setup slacks	-319.7300 ns	Zero means no negative setup slack was summed in the reported set.
Minimum reported hold slack	0.8600 ns	Positive in the supplied summary; no negative hold paths are reported.
Negative setup paths in reported set	100 of 100	Count copied from the supplied summary.
Negative hold paths in reported set	0 of 100	Count copied from the supplied summary.

Corner metrics



Assessment

Setup timing is violated in the reported path set. Review the critical paths and constraints for this corner; do not treat the positive hold slack as evidence that setup timing is acceptable.

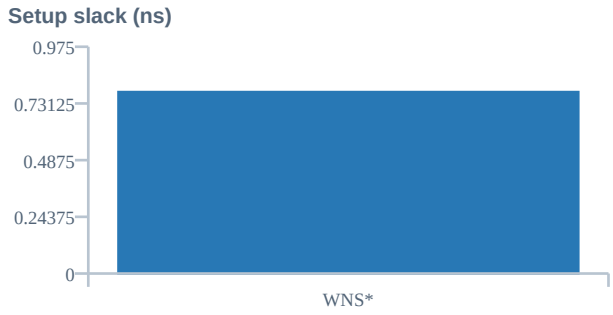
ss_100C_1v60

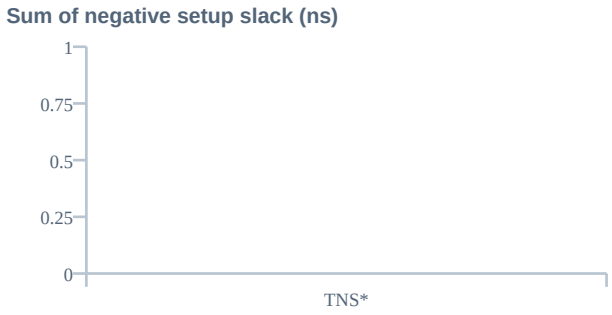
Process: SS | Temperature: 100 °C | Voltage: 1.60 V

MET

Metric	Reported value	Interpretation
WNS* / minimum reported setup slack	0.7800 ns	Non-negative for the reported setup path set.
TNS* / sum of negative reported setup slacks	0.0000 ns	Zero means no negative setup slack was summed in the reported set.
Minimum reported hold slack	0.6100 ns	Positive in the supplied summary; no negative hold paths are reported.
Negative setup paths in reported set	0 of 100	Count copied from the supplied summary.
Negative hold paths in reported set	0 of 100	Count copied from the supplied summary.

Corner metrics





Assessment

The supplied summary marks setup timing MET for the reported path set and shows positive minimum hold slack. This is not a substitute for confirming complete design-wide STA summaries.

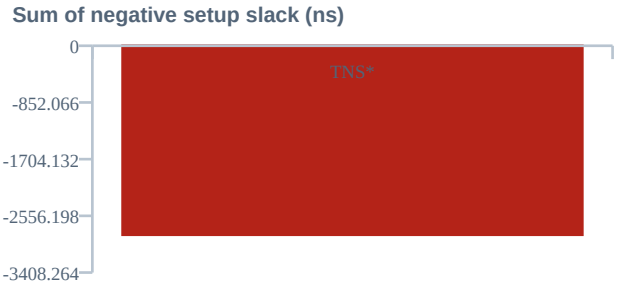
ss_n40C_1v28

Process: SS | Temperature: -40 °C | Voltage: 1.28 V

VIOLATED

Metric	Reported value	Interpretation
WNS* / minimum reported setup slack	-32.4800 ns	Negative means a setup violation exists in the reported path set.
TNS* / sum of negative reported setup slacks	-2840.2200 ns	Zero means no negative setup slack was summed in the reported set.
Minimum reported hold slack	1.3900 ns	Positive in the supplied summary; no negative hold paths are reported.
Negative setup paths in reported set	100 of 100	Count copied from the supplied summary.
Negative hold paths in reported set	0 of 100	Count copied from the supplied summary.

Corner metrics



Assessment

Setup timing is violated in the reported path set. Review the critical paths and constraints for this corner; do not treat the positive hold slack as evidence that setup timing is acceptable.

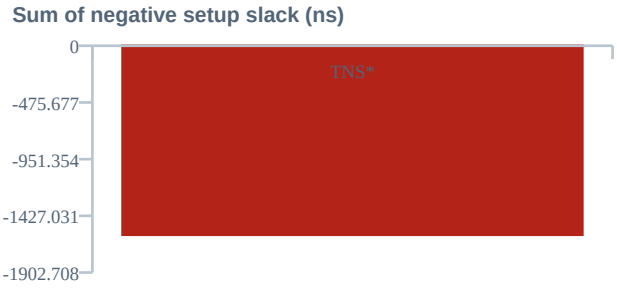
ss_n40C_1v35

Process: SS | Temperature: -40 °C | Voltage: 1.35 V

VIOLATED

Metric	Reported value	Interpretation
WNS* / minimum reported setup slack	-18.4600 ns	Negative means a setup violation exists in the reported path set.
TNS* / sum of negative reported setup slacks	-1585.5900 ns	Zero means no negative setup slack was summed in the reported set.
Minimum reported hold slack	1.0800 ns	Positive in the supplied summary; no negative hold paths are reported.
Negative setup paths in reported set	100 of 100	Count copied from the supplied summary.
Negative hold paths in reported set	0 of 100	Count copied from the supplied summary.

Corner metrics



Assessment

Setup timing is violated in the reported path set. Review the critical paths and constraints for this corner; do not treat the positive hold slack as evidence that setup timing is acceptable.

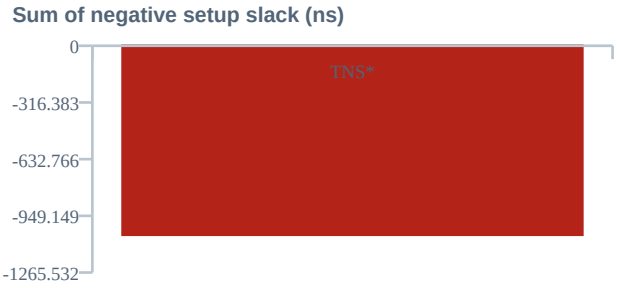
ss_n40C_1v40

Process: SS | Temperature: -40 °C | Voltage: 1.40 V

VIOLATED

Metric	Reported value	Interpretation
WNS* / minimum reported setup slack	-12.6100 ns	Negative means a setup violation exists in the reported path set.
TNS* / sum of negative reported setup slacks	-1054.6100 ns	Zero means no negative setup slack was summed in the reported set.
Minimum reported hold slack	0.9200 ns	Positive in the supplied summary; no negative hold paths are reported.
Negative setup paths in reported set	100 of 100	Count copied from the supplied summary.
Negative hold paths in reported set	0 of 100	Count copied from the supplied summary.

Corner metrics



Assessment

Setup timing is violated in the reported path set. Review the critical paths and constraints for this corner; do not treat the positive hold slack as evidence that setup timing is acceptable.

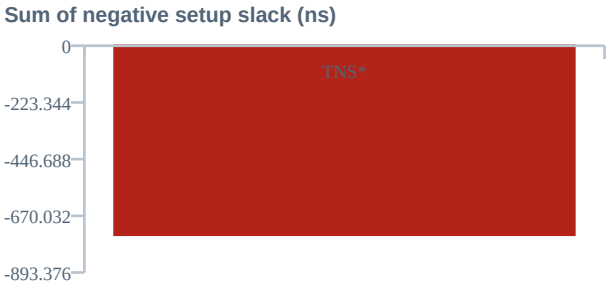
ss_n40C_1v44

Process: SS | Temperature: -40 °C | Voltage: 1.44 V

VIOLATED

Metric	Reported value	Interpretation
WNS* / minimum reported setup slack	-9.2100 ns	Negative means a setup violation exists in the reported path set.
TNS* / sum of negative reported setup slacks	-744.4800 ns	Zero means no negative setup slack was summed in the reported set.
Minimum reported hold slack	0.8200 ns	Positive in the supplied summary; no negative hold paths are reported.
Negative setup paths in reported set	100 of 100	Count copied from the supplied summary.
Negative hold paths in reported set	0 of 100	Count copied from the supplied summary.

Corner metrics



Assessment

Setup timing is violated in the reported path set. Review the critical paths and constraints for this corner; do not treat the positive hold slack as evidence that setup timing is acceptable.

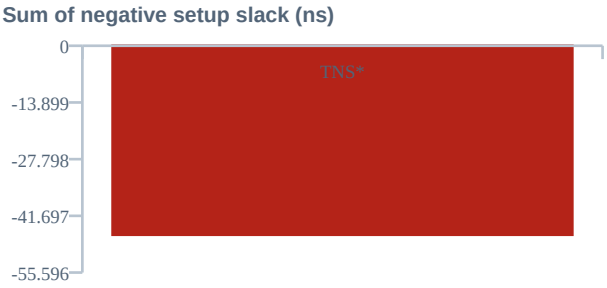
ss_n40C_1v60

Process: SS | Temperature: -40 °C | Voltage: 1.60 V

VIOLATED

Metric	Reported value	Interpretation
WNS* / minimum reported setup slack	-1.4000 ns	Negative means a setup violation exists in the reported path set.
TNS* / sum of negative reported setup slacks	-46.3300 ns	Zero means no negative setup slack was summed in the reported set.
Minimum reported hold slack	0.5700 ns	Positive in the supplied summary; no negative hold paths are reported.
Negative setup paths in reported set	63 of 100	Count copied from the supplied summary.
Negative hold paths in reported set	0 of 100	Count copied from the supplied summary.

Corner metrics



Assessment

Setup timing is violated in the reported path set. Review the critical paths and constraints for this corner; do not treat the positive hold slack as evidence that setup timing is acceptable.

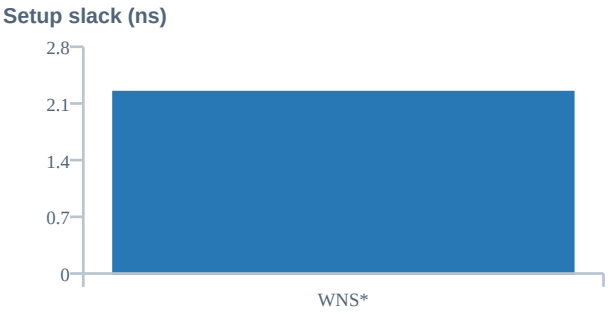
ss_n40C_1v76

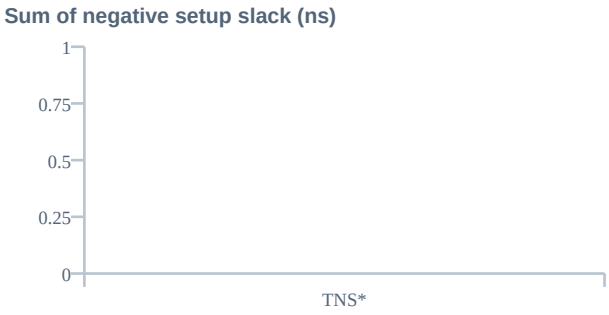
Process: SS | Temperature: -40 °C | Voltage: 1.76 V

MET

Metric	Reported value	Interpretation
WNS* / minimum reported setup slack	2.2400 ns	Non-negative for the reported setup path set.
TNS* / sum of negative reported setup slacks	0.0000 ns	Zero means no negative setup slack was summed in the reported set.
Minimum reported hold slack	0.4400 ns	Positive in the supplied summary; no negative hold paths are reported.
Negative setup paths in reported set	0 of 100	Count copied from the supplied summary.
Negative hold paths in reported set	0 of 100	Count copied from the supplied summary.

Corner metrics





Assessment

The supplied summary marks setup timing MET for the reported path set and shows positive minimum hold slack. This is not a substitute for confirming complete design-wide STA summaries.

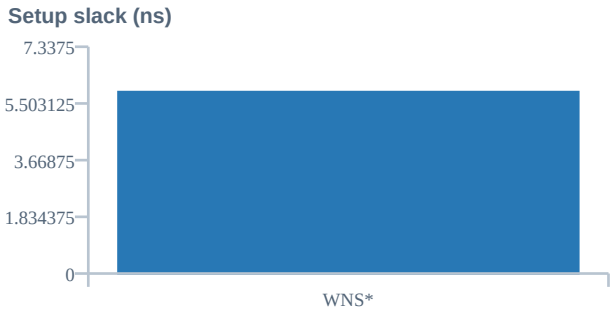
tt_025C_1v80

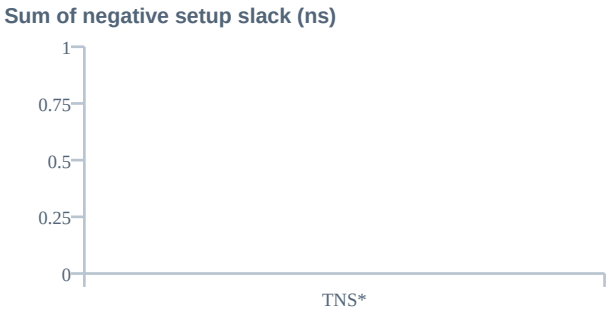
Process: TT | Temperature: 25 °C | Voltage: 1.80 V

MET

Metric	Reported value	Interpretation
WNS* / minimum reported setup slack	5.8700 ns	Non-negative for the reported setup path set.
TNS* / sum of negative reported setup slacks	0.0000 ns	Zero means no negative setup slack was summed in the reported set.
Minimum reported hold slack	0.3000 ns	Positive in the supplied summary; no negative hold paths are reported.
Negative setup paths in reported set	0 of 100	Count copied from the supplied summary.
Negative hold paths in reported set	0 of 100	Count copied from the supplied summary.

Corner metrics





Assessment

The supplied summary marks setup timing MET for the reported path set and shows positive minimum hold slack. This is not a substitute for confirming complete design-wide STA summaries.

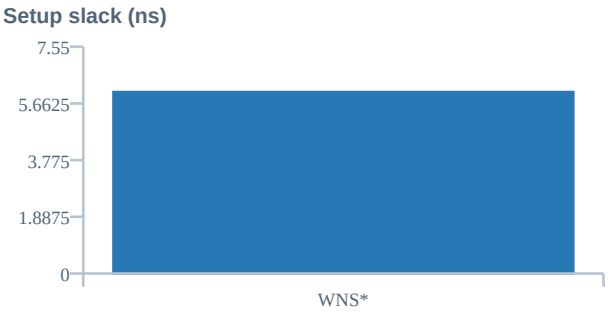
tt_100C_1v80

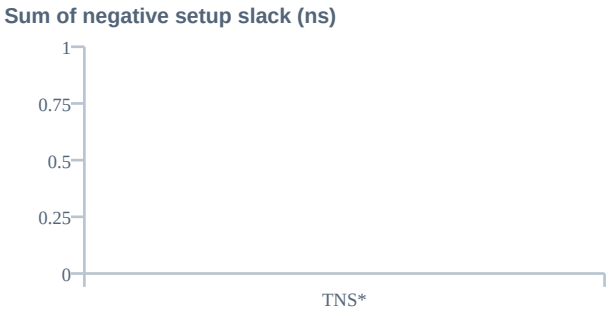
Process: TT | Temperature: 100 °C | Voltage: 1.80 V

MET

Metric	Reported value	Interpretation
WNS* / minimum reported setup slack	6.0400 ns	Non-negative for the reported setup path set.
TNS* / sum of negative reported setup slacks	0.0000 ns	Zero means no negative setup slack was summed in the reported set.
Minimum reported hold slack	0.3100 ns	Positive in the supplied summary; no negative hold paths are reported.
Negative setup paths in reported set	0 of 100	Count copied from the supplied summary.
Negative hold paths in reported set	0 of 100	Count copied from the supplied summary.

Corner metrics





Assessment

The supplied summary marks setup timing MET for the reported path set and shows positive minimum hold slack. This is not a substitute for confirming complete design-wide STA summaries.